# Fraudulent Bitcoin Transaction Detection

*A drift-aware, graph-stacked ensemble for the Elliptic transaction graph*

Code, data, and full pipeline: github.com/yushanedi/bt4012-bitcoin-model-edison

## 1. Introduction

Bitcoin's pseudonymous ledger carries ransomware payouts, darknet-market proceeds and laundering alongside ordinary payments, so classifiers that flag suspicious transactions from their attributes and position in the payment graph are a core fraud-analytics tool for compliance teams and investigators. The task is to classify **157,101** transactions in an Elliptic-style graph (**234,355** directed edges, 49 `time_steps`) as illicit (1) or licit (0) from 165 anonymized features (94 local, 71 neighbourhood aggregates). `train.csv` covers `time_steps` 1–35 with only **31,235** labeled rows (11.7% illicit); `test.csv` covers 36–49 and is scored by ROC-AUC. Labels are scarce and imbalanced, the scoring period lies entirely in the future of the training data, and fraud patterns drift over time.

Our approach had three strands: **(1)** validation that mimics the forward-in-time gap between training and scoring; **(2)** drift-aware and graph-structural feature engineering; and **(3)** a final ensemble adding neighbour-score stacking, unlabeled transactions as weak licit examples, and rank-averaging across feature views and tree models. The submitted ensemble scored a ROC-AUC of **0.96758**.

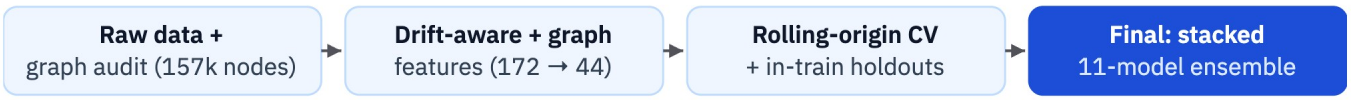



*Diagram 1. End-to-end pipeline.*

## 2. Methodology

### 2.1 Data Preprocessing and Feature Engineering

**Cleaning.** No feature values are missing, so no imputation was needed. The 110,537 unlabeled `train.csv` rows (78%) were never given imputed labels: baselines use labeled rows only, and ensemble members add unlabeled rows as low-weight licit examples (§2.3). 58,437 of the edges have an endpoint that is in neither file.

**Graph features.** Every edge between known transactions joins two nodes in the *same* `time_step`, so graph information never flows between periods. Within each `time_step` we computed 7 topology features: in-, out- and total degree, PageRank, clustering coefficient, k-core number and component size (172 features in total). The labeled graph is strongly homophilous: 94.7% of labeled edges join same-class transactions, and an illicit node's neighbour is illicit 57.4% of the time, against 2.8% for a licit node.

**Drift-aware transformation.** A classifier separating `train.csv` from `test.csv` rows on features alone reached ROC-AUC ≈1.00, against ≈0.50 on a random split of `train.csv`, confirming a real shift between periods. Of the 30 features that most separate the periods, the 13 that are also in the top 60 for label importance (e.g. `feat_2`) were replaced by their

**percentile rank within their own `time_step`**; the other 16 were dropped, and the remaining 143 were kept raw. Within-step ranks of all 172 features also form an alternative view.

**Feature selection.** Sweeping the number of top gradient-boosted-tree importance features from 20 to 156 under rolling-origin CV (§2.2) found that a compact **top-44** subset generalized best; weaker features mostly added period-specific noise.

### 2.2 Model Development and Validation

**Models.** Logistic regression on standardized features; GCN, GAT and GIN (PyTorch Geometric, trained transductively on the full graph with class-weighted cross-entropy and early stopping); and tree ensembles (XGBoost, CatBoost, Random Forest, ExtraTrees) plus a small MLP.

**Validation. (i) Near-split:** train on `time_step` < 30, validate on 30–35, used to compare baselines. **(ii) Rolling-origin CV:** four splits within `train.csv`, each with a 7–13-step gap (train ≤15 → validate 22–27, ≤18 → 25–30, ≤22 → 29–34, ≤20 → 33–35). This scheme drove feature selection and a random hyperparameter search (learning rate, depth, min-child-weight, L1/L2, subsampling). **(iii) In-train holdouts:** the full final pipeline re-run with a later slice treated as unlabeled test data (train ≤22 → score 29–35; train ≤25 → score 30–35).

**Imbalance and overfitting.** Boosted models set `scale_pos_weight` to the negative:positive count of their training rows (7.6 on labeled rows alone, higher when unlabeled rows are added). The GNNs use inverse-frequency class weights, and the bagged trees use balanced subsample weights. Overfitting is limited by feature reduction, L2 regularization and column subsampling, early stopping and seed averaging. Stacked features are built **out-of-fold** over five blocks of `time_steps`, so no training row's neighbour features come from a model that saw its own label.

### 2.3 Model Exploration and Final Strategy

Under the near-split, gradient boosting clearly beat the GNNs and logistic regression (Table 2). This fits the graph audit: message passing only smooths within a period, while learned thresholds transfer across periods. Several shift-targeted methods, all evaluated with rolling-origin CV, were tried and dropped: CORAL and importance reweighting (lower CV), domain-adversarial and entropy-minimizing networks (below boosting), label propagation and self-training (weak, or hurt), period-specific mixtures of experts (hurt), and smoothing scores over the graph (CV 0.924 → 0.81–0.86).

**Final ensemble.** The submission extends a single XGBoost (learning rate 0.02, depth 6, min-child-weight 5, L2 0.5, column sample 0.6, 800 trees) in three ways. **Weak unlabeled negatives:** the unlabeled training rows enter as licit with weight 0.05–0.15, widening the model's view of ordinary activity. **Neighbour-score stacking:** a first-stage model scores every node, each node gains the mean and maximum of its in-, out- and undirected neighbours' scores, and the next stage trains on them, for up to three stages. Edges leaving the dataset are attached to one shared placeholder node, so these features also flag transactions that link outside it. **Rank averaging:** 11 models, each scoring a different feature view, are converted to ranks and averaged with fixed integer weights (Table 1).

| Member(s) | Model | Unlabeled weight | Stages | Weight |
|---|---|---|---|---|
| Core blend: top-44 (×3), top-44 → top-44+ranks, local ranks | XGBoost, 3 seeds | 0.05 / 0.05 / 0 | 3 / 2 / 1 | 1 |
| Top-44 | XGBoost / ExtraTrees / Random Forest | 0.15 / 0.05 / 0 | 3 | 3 / 2 / 1 |
| Top-44 + all within-step ranks | XGBoost | 0.15 | 3 | 1 |
| Neighbourhood aggregates (feat 95–165) | XGBoost ×3 | 0 / 0.05 / 0.15 | 3 | 1 each |
| Local-feature ranks (feat 1–94) | XGBoost | 0.15 | 3 | 1 |

*Table 1. Final ensemble members (11 models), rank-averaged with the weights shown.*

**Selection.** We submitted this ensemble as our final model because it achieved our highest submitted ROC-AUC, **0.96758**. The single top-44 XGBoost is kept in the notebook as a reference model.

## 3. Results and Discussion

| Baseline (near-split) | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.693 | 0.340 | 0.983 | 0.505 | 0.916 |
| GCN | 0.875 | 0.565 | 0.938 | 0.705 | 0.959 |
| GAT | 0.491 | 0.238 | 0.994 | 0.384 | 0.950 |
| GIN | 0.898 | 0.970 | 0.371 | 0.537 | 0.950 |
| **XGBoost (raw features)** | **0.986** | **0.955** | **0.957** | **0.956** | **0.997** |

*Table 2. Baselines on the near-split (4,854 validation rows); thresholded metrics at 0.5.*

**Validation design matters as much as the model.** The near-split nearly saturates (XGBoost 0.997) because validation starts right after the training cutoff. Rolling-origin CV with a 7–13-step gap puts the 156-feature drift-corrected model at ≈0.89 and the top-44 XGBoost at ≈0.92, so choosing features and hyperparameters on the near-split would have been badly misleading.

| Pipeline (in-train holdout) | ≤22 → 29–35 | ≤25 → 30–35 | Mean |
|---|---|---|---|
| **Final ensemble (Table 1)** | **0.791** | **0.993** | **0.892** |

*Table 3. ROC-AUC with the full pipeline re-run on a later slice of `train.csv` treated as unlabeled. Submitted, the final ensemble scored **0.96758**.*

**Possible causes of model behaviour.** The same pipeline ranges from 0.79 to 0.99 across holdouts, so how well a model transfers depends heavily on which period is scored. For the ensemble, plausible but unconfirmed mechanisms are that neighbour scores exploit within-period homophily, that weak negatives widen the model's picture of normal activity, and that placeholder links flag transactions connected outside the dataset.

**Strengths:** no model is trained on test labels, validation reproduces the forward-in-time setting, stacking is out-of-fold, and the full pipeline is re-run end to end on in-train holdouts. **Limitations:** on those holdouts the final ensemble did not outperform simpler single-model pipelines, so its submitted score may not carry over to other periods. The holdouts are noisy, the ensemble is slower (about 12 minutes) and harder to interpret than one model, and the placeholder node mixes a data artifact into the neighbour features.

## 4. Conclusion and Reflection

We built a pipeline for forward-in-time fraud detection on a transaction graph: a graph audit, drift-aware features reduced to a top-44 core, rolling-origin validation, and a final 11-model stacked ensemble that scored a ROC-AUC of **0.96758** on submission. The clearest finding is about evaluation: an adjacent split reports a near-perfect 0.997, while gap-matched validation puts the same family of models at about 0.89–0.94.

**What worked:** rolling-origin validation for model choices, within-period rank normalization of drifting but informative features, aggressive feature reduction, and gradient-boosted trees over GNNs on a graph whose edges never cross periods. **What could be improved:** our holdouts did not track the submitted score, so a better proxy for the scoring period, such as more holdout slices or weighting toward recent periods, is the key next step, along with a deliberate treatment of off-dataset edges.

**Reflection:** AI-driven fraud detection is a moving-target problem. Both fraud and legitimate activity shift over time, so how a model is validated decides how far its score can be trusted. A model that looks better on one period may not be better on the next. Validation that mirrors deployment, honest ablations, and stating plainly where the evidence is weak matter as much as architecture.